

SERVICENOW CERTIFIED APPLICATION DEVELOPER

SmartInternz

Project Tittle

Automated Network Request Management in ServiceNow

Platform	ServiceNow
Application	Service Catalog and Flow Designer
Primary Catalog Item	Network Request
Backend table	Network DataBase Table
Automation	Flow Designer
Document Type	Project implementation Report

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1. Project Overview

The **Automated Network Request Management in ServiceNow** project automates network request submission, approval, email notifications, and record updates using **Service Catalog and Flow Designer**. It reduces manual work and makes request management faster and more efficient.

1.2. Project Objective

The **main objective** of the project is to **automate the complete network request process in ServiceNow**, from request submission to approval, email notification, and record update, reducing manual work and improving efficiency.

- To automate the **network request submission process**.
- To create and manage network requests using **ServiceNow Service Catalog**.
- To automate the **approval process** using Flow Designer.
- To send **automatic email notifications** to users and approvers.
- To automatically **update request records and statuses**.
- To reduce **manual work and processing time**.
- To improve **request tracking and management**.

1.3 Request Workflow

- **User submits** a Network Request through the Service Catalog.
- The request details are **stored in the Network Database Table**.
- **Flow Designer** automatically starts the workflow.
- The request is sent to the **approver**.
- The **approves or rejects** the request.
- An **email notification** is sent based on the request status.
- The system **updates the request record and status** automatically.
- The request is **completed and tracked** in ServiceNow.

1.4: End-to-End Process

User Request → Catalog Form → Database Record → Flow Trigger → Approval → Email Notification → Record Update → Completion

- **Submit Request** – User opens the **Network Request** catalog item and fills in the required details.
- **Create Request** – The submitted information is stored in the **Network Database Table**.
- **Start Flow** – Flow Designer automatically triggers the network request workflow.
- **Send for Approval** – The request is sent to the assigned approver.
- **Approval Decision** – The
Proper approves or rejects the request.
- **Send Email** – The system automatically sends an email notification about the request status.

- **Update Record** – The Network Database record is updated with the latest status.
- **Complete Request** – The request reaches its final status and can be tracked in ServiceNow.

2. ServiceNow Start Building Page: -

The **Automated Network Request Management in ServiceNow** is a project developed on the ServiceNow platform to automate and simplify the handling of network-related service requests within an organization. The project begins in the ServiceNow developer environment. The Start Build page provides access to application-development capabilities, while the ServiceNow home page and application navigator provide access to configuration areas used during implementation.

✅ **Entry Page:** The ServiceNow entry page acts as the starting interface where users can submit, track, and manage requests.

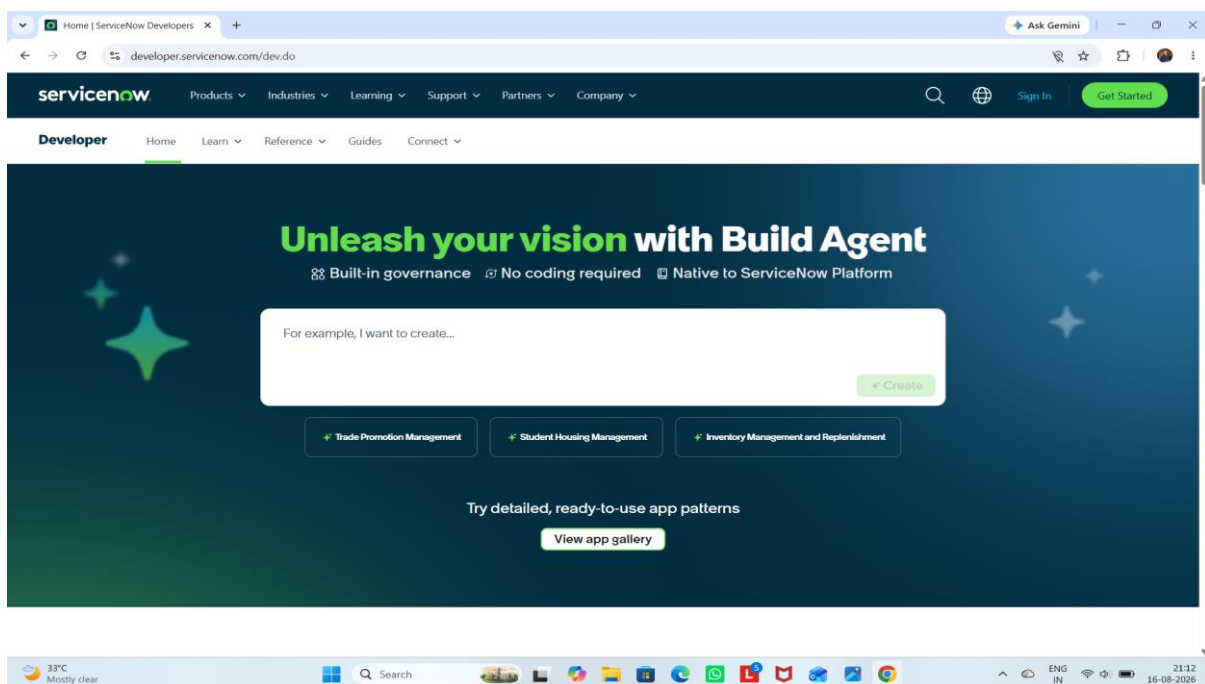


Fig. 1: ServiceNow Developer- Start Building page

3. ServiceNow Home / Application Navigation: -

- Open the ServiceNow Developer instance.
- Open the start Building page.
- Go to the ServiceNow home page.
- Click All in the Application Navigator.
- Use the search box to find the required module.

Key Activities: -

- Open the ServiceNow Developer instance.
- From the Start Building page, open the required development / configuration area.
- Click All and search for the required module in the Application Navigator.
- Navigate to All>service Catalog>Catalog Definitions>Maintain items.
- Go to the Catalog Item Network Request page.

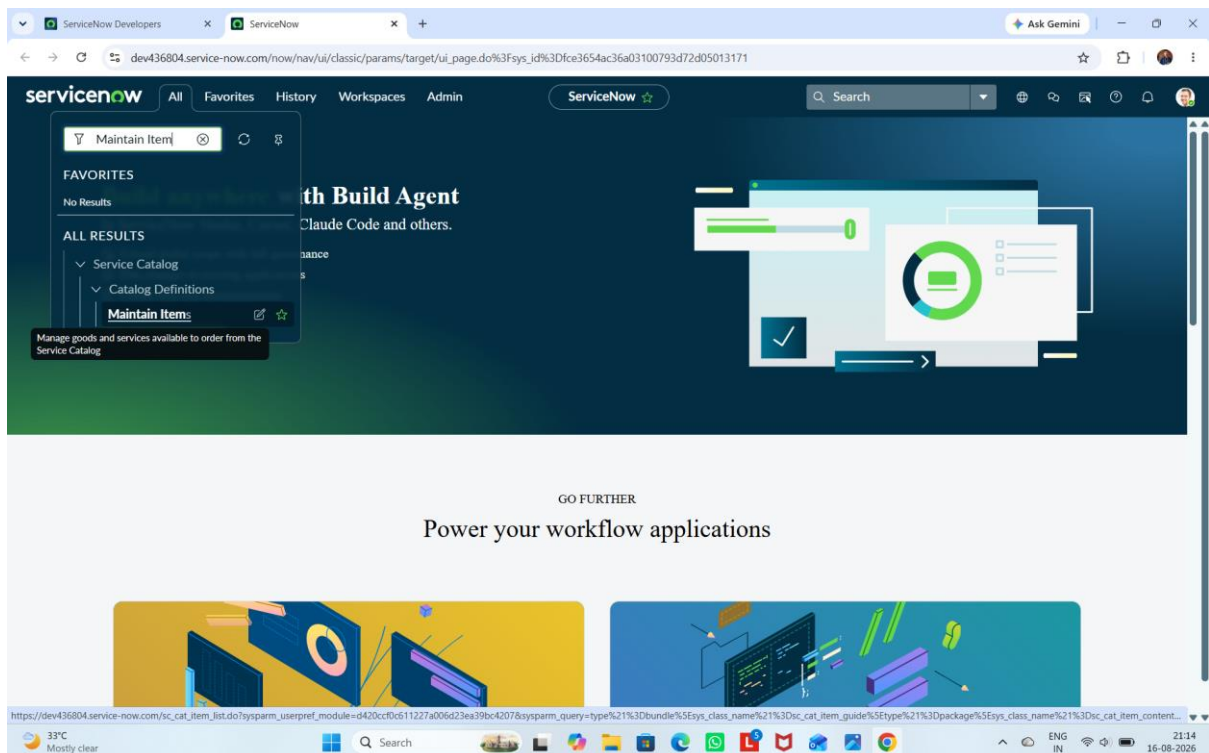


Fig.2: ServiceNow Home page and application Navigation

4.1 Creation of Catalog Item: -

✅ Navigate to all> service Catalog>Catalog definitions>maintain Items. Search for Network Request and open the catalog item. This is a parent record for Variables, Variable Sets, and Catalog UI Policies.

servicenow All Favorites History Workspaces Catalog Item - Network Request

Search

Copy Try It Update Edit in Catalog Builder Delete

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name

Application

Catalogs

Category

State

Checked out

Owner

Secondary owners

Active ☒

Fulfillment automation level

Fig.3: Network Request Catalog Item Configuration

4.2. Configuration Steps: -

- ★ Navigate to all>Service Catalog>Catalog Definitions>Maintain Items.
- ★ Search for Network Request and open it. If it does not exist, click New and create it.
- ★ Enter the required Catalog item properties such as Name, Catalog, Category, Description, and other project-required values.
- ★ Click Submit/Update to save the catalog item.
- ★ Scroll to the related lists and configure Variables, Variable Sets, and Catalog UI Policies.

4.3. Variable Configuration (10 – Variables)

- Click All.
- Go to Service Catalog → Catalog Definitions → Maintain Items.
- Open Network Request.
- Scroll to the Variables related list.
- Click New.
- Select the required Variable Type.
- Enter the Order number.
- Enter the Question / Label.
- Enter the variable Name.
- Configure any additional properties required.
- Click Submit.
- Repeat the same steps until all 10 variables are created.

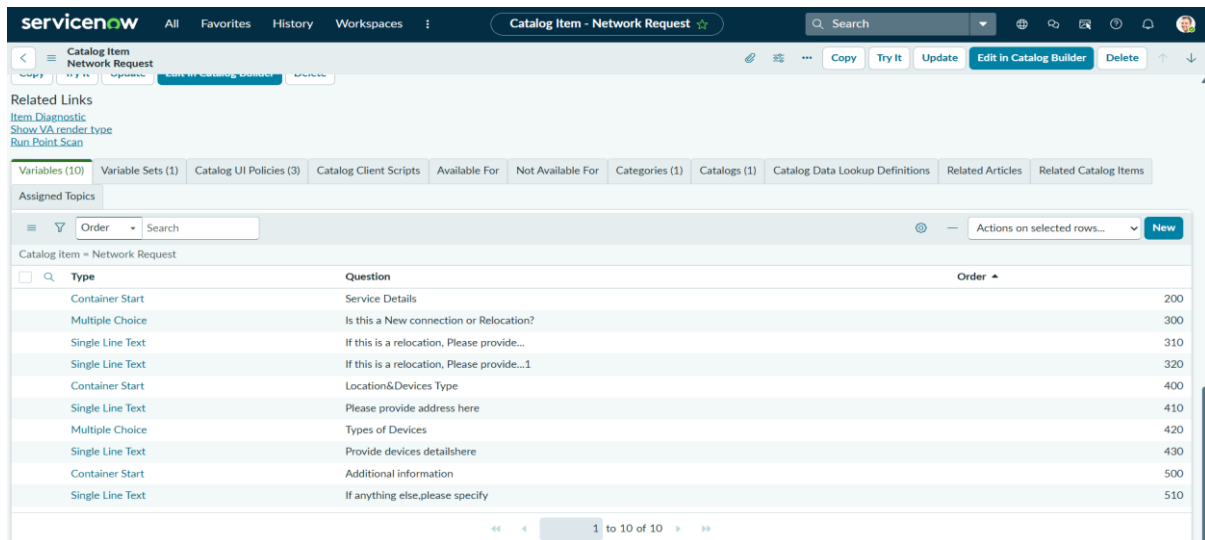


Fig.4: Network Request – Variable related list (10)

4.4. Variable Configuration Principles: -

- ★ Select the required variable Type, such as Single Line Text, Multiple Choice, Reference, or Attachment.
- ★ Enter the Order Value, so the variables appear in the required sequence.
- ★ Enter the Question/Label displayed to the requester.
- ★ Enter a meaningful Name for later Flow Designer mapping.
- ★ Use container variables where required to group related fields.

4.5. Variable Groups and Purpose: -

Variable Groups	Purpose
Connection type	Captures whether the request is a new connection or Relocation.
Relocation details	Captures information required for a relocation request.
Location / address	Captures the address associated with the request.
Device information	Captures device type and device specific details.
Additional information	Capture any remaining information required to process the request.

5.1. Catalog Item Network Request – Variable sets

✓ Open All > Service Catalog > Catalog Definitions > Maintain Items > Network Request. Scroll to Variable Sets and click New. Create Requester information, set its order, and click Submit. Open it and configure its variables.

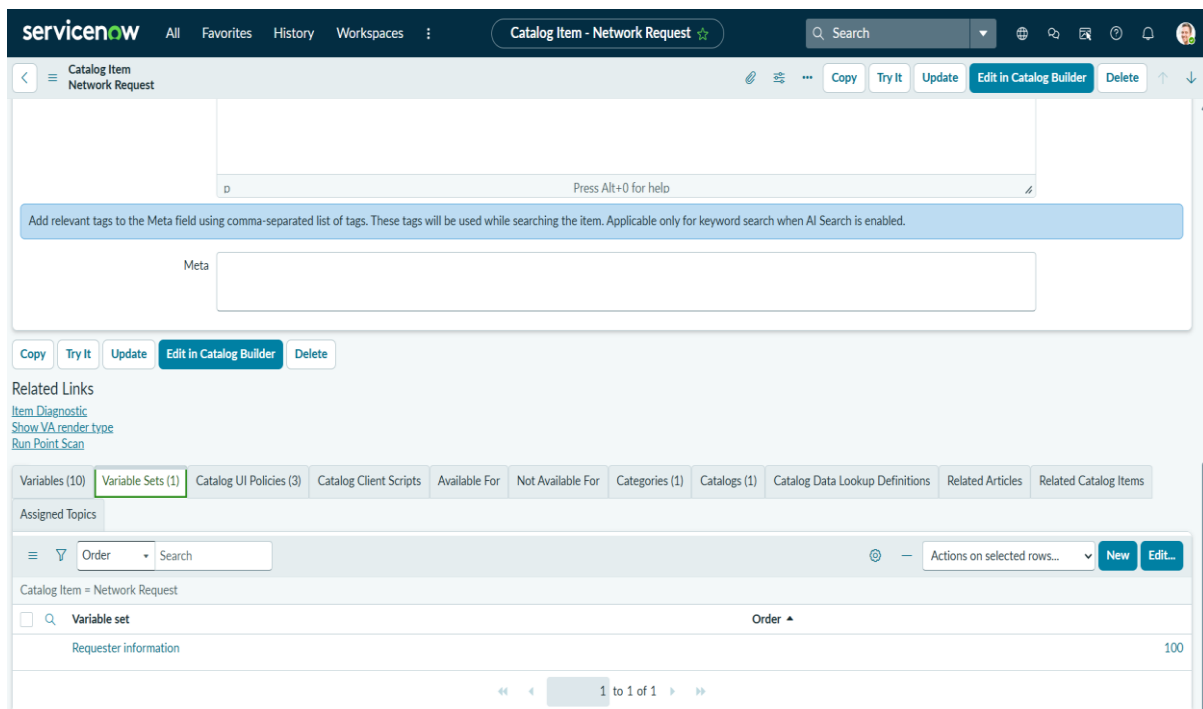


Fig. 5: Network Request – Variable Sets

6.1. Request Information: -

The Requester information Variable Set contains five variables. Keeping these fields in a Variable Set makes the requester information reusable and logically separates it from network-request-specific data.

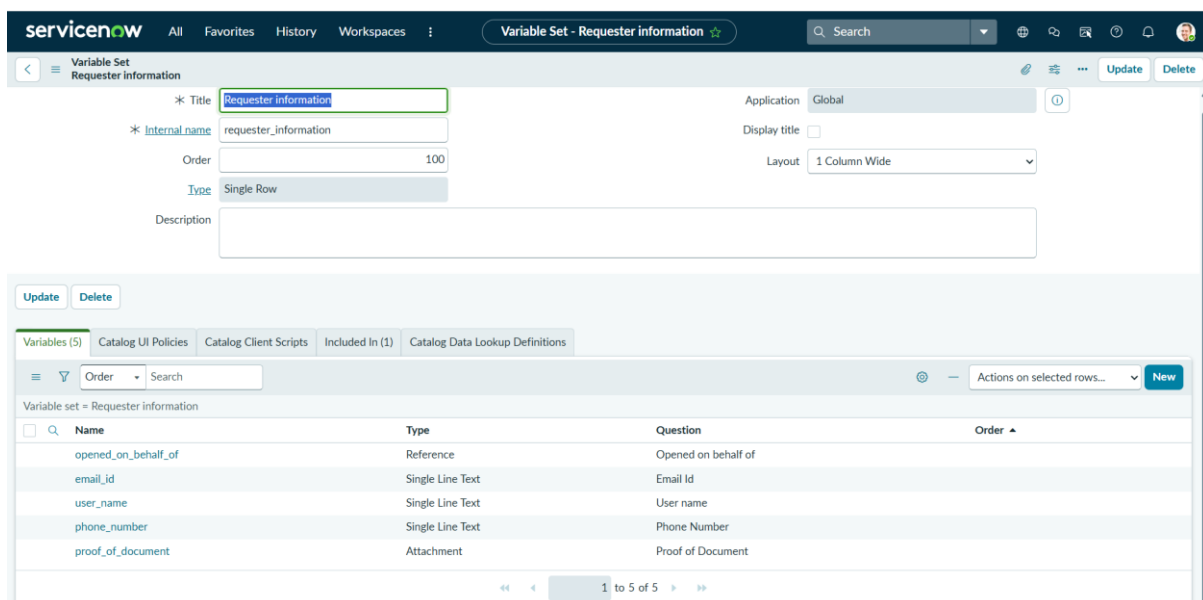


Fig.5: Request Information (Variable Sets)

6.2. Explanation of variables in Request Information: -

Variables	Purpose/Type
Email_id	Capture the requester's email address.

proof_of_document	Provides an attachment field for supporting documentation.
opened_on_behalf_of	References the user on whose behalf the request is opened.
username	Captures the requester's username.
phonenumber	Capture the requester's contact number.

7.1. Catalog UI Policies: -

- Click All.
- Go to Service Catalog → Catalog Definitions → Maintain Items.
- Open Network Request.
- Scroll to Catalog UI Policies.
- Click New.
- Enter the UI Policy name.
- Select the required Catalog Item condition.
- Configure the UI Policy Actions.
- Click Submit / Update.
- Repeat the same process for the second Catalog UI Policy.
- Test the form to verify the field Show or hide correctly.

The screenshot displays the ServiceNow interface for configuring a Catalog UI Policy. The top navigation bar shows 'servicenow' and various tabs like 'All', 'Favorites', 'History', and 'Workspaces'. The main header is 'Catalog Item - Network Request'. Below this, there's a search bar and a 'Meta' field with a placeholder 'Press Alt+0 for help'. A blue banner提示 says: 'Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Applicable only for keyword search when AI Search is enabled.' Below the Meta field are buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. The 'Related Links' section includes 'Item Diagnostic', 'Show VA render type', and 'Run Point Scan'. The 'Assigned Topics' section shows a table with columns: 'Short description', 'Variable set', 'Conditions', 'Reverse if false', 'On load', 'Inherit', 'Updated', and 'Order'. The table contains three rows of data.

Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
Types of devices is others	(empty)		true	true	false	2026-08-11 01:59:36	100
Types of devices others	(empty)		true	true	false	2026-08-12 10:42:53	100
Relocation fields hiding	(empty)		true	true	false	2026-08-11 02:00:20	100

Fig.4.4: Network Request –Catalog UI Policies

7.2. Policies Configuration: -

Policy	Purpose
Types of devices is others	Controls the behavior of additional device information when the requester selects Others.
Relocation fields hiding	Controls relocation-related fields based on the connection type selection.

7.3. Explanation: -

- ❖ The policy “**Types of devices is others**” manages additional device information fields when the requester selects *Others*.
- ❖ The policy “**Relocation fields hiding**” hides relocation-related fields depending on the selected connection type.
- ❖ These configurations ensure that only relevant fields are displayed, improving form clarity and user experience.
- ❖ Policies help automate dynamic form behavior without requiring scripting, making the catalog item more interactive and efficient.
- ❖ These points summarize how **Catalog UI Policies** are configured in your **Automated Network Request Management** project to control form logic and enhance workflow automation.

8.1. Table Creation: -

- Click All in the Application Navigator.
- Go to System Definition → Tables.
- Click New.
- In Label, enter Network Database Table.
- Verify the generated Name.
- Keep Application as Global if required by the project.
- Keep Create module selected if required.
- Enter the New menu name as Network Database Table if required.
- Review the table configuration.
- Click Submit.

Name	Extends table	Extensible	Updated
account_subscription_entitlement	(empty)	false	2026-04-30 01:10:38
adaptive_auth_event	(empty)	false	2026-04-30 01:00:31
agent_assist_recommendation	Application File	false	2026-04-30 00:55:26
agent_file	(empty)	false	2026-04-30 00:42:28
aisa_rp_config	Application File	false	2026-04-30 00:58:38
aisa_ui_action	Application File	false	2026-04-30 00:58:38
ais_acl_overrides	Application File	false	2026-04-30 00:23:37
ais_active_table_ingestion_tracker	(empty)	false	2026-04-30 00:23:35
ais_api_transaction_log	(empty)	false	2026-04-30 00:23:39
ais_async_genius_result	(empty)	false	2026-04-30 00:23:40
ais_async_request	(empty)	false	2026-04-30 00:23:40
ais_child_table	Application File	false	2026-04-30 00:23:38
ais_configuration_attribute	(empty)	false	2026-04-30 00:23:35
ais_connection	(empty)	false	2026-04-30 00:23:40
ais_country_to_search_language	Application File	false	2026-04-30 00:23:40
ais_custom_matcher	Application File	false	2026-04-30 00:23:38
ais_datasource	Application File	false	2026-04-30 00:23:39
ais_datasource_attribute	Application File	false	2026-04-30 00:23:35
ais_datasource_field_attribute	Application File	false	2026-04-30 00:23:36

Fig. 8. ServiceNow Table Configuration / Navigation

8.2: Create the Network Database Table: -

Create the Network Database Table in the ServiceNow project is:

- Go to **System Definition** → **Tables** in the ServiceNow navigation panel.
- Click **New** to create a new table for storing network request data.
- Enter a **Label** (e.g., *Network Request Database*) and a **Name** (auto-generated).
- Add **Columns (Fields)** such as *Request ID*, *Requester Name*, *Device Type*, *Location*, *Status*, and *Date*.
- Define **Data Types** for each field — text, choice, date/time, or reference.
- Set **Access Controls** to manage who can view or modify records.
- Save the table and verify it appears under the *Network Request* application.
- Link the table to your **Catalog Item** or **Workflow**, so submitted requests are stored automatically.
- These points describe the basic process of creating and configuring a **Network Database Table** for the *Automated Network Request Management* project in ServiceNow.

The screenshot shows the ServiceNow 'Tables' configuration page for a table named 'Network DataBase Table'. The page has a dark blue header with the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, Admin, and Tables (which is highlighted). A search bar is located on the right side of the header. Below the header, the table name 'Network DataBase Table' is displayed. The configuration form includes several fields: 'Label' (Network DataBase Table), 'Name' (u_network_database_table), and 'Extends table' (empty). On the right side, there are checkboxes for 'Create module' and 'Create mobile module', both of which are checked. Below these, there is a dropdown menu for 'Add module to menu' with the option '-- Create new --' selected, and a text field for 'New menu name'. At the bottom right, there is a checkbox for 'Remote Table' which is unchecked. The page also features buttons for 'Delete', 'Update', and 'Delete All Records'.

Fig.9: Creation of Network DataBase Table

After the table is created, the Tables list can be used to locate and manage the configured table. This page also demonstrates the ServiceNow table-management environment used during the project.

8.3. Purpose: -

- The table stores all submitted **network request records** in a structured format.
- It acts as the **central database** for tracking request details, status, and user information.
- Each record includes fields like *Request ID*, *Requester Name*, *Device Type*, *Location*, and *Status*.
- Helps in **workflow automation** by linking catalog items and approvals to Stored
- Enables **reporting and analytics** to measure performance and request trends.
- Maintains **data consistency and accessibility** for IT teams managing network operations.

9.1. How to Create the Fields: -

- Click All.
- Go to System Definition → Tables. Automated Network Request Management in ServiceNow
- Search for a Network Database Table.
- Open the table.
- Open the Columns / Fields related list.
- Click New.
- Select the field Type.
- Enter the Column label / Name.
- Configure Reference, Max length, Default value, Display, or other required properties.
- Click Submit.
- Repeat for all required fields.
- Verify that all fields are listed in the table.

Column label	Type	Reference	Max length	Default value	Display
Assigned to	Reference	User	32		false
Requested For	String	(empty)	40		false
Updated by	String	(empty)	40		false
Approval	String	(empty)	40		false
Request Number	String	(empty)	40		false
Created by	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Work Status	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Approval history	String	(empty)	40		false
Assignment Group	Reference	Group	32		false
Updates	Integer	(empty)	40		false
Date of Enquiry	Date	(empty)	40		false
Device Details	String	(empty)	40		false
Customer Address	String	(empty)	40		false

Fig.11: Network DataBase Table – Columns and fields

8.2. Table: -

Field	Description
Assigned to	Reference field used to identify the assigned user.
Customer Address	Stores the customer's address information.
Requested For	Stores the requested-for information.
Work Status	Stores the current work status of the request.
Device Details	Stores device-related information.
Customer Document	Store Customer Document information.
Request Number	Stores the identifier used to track the request.
Assignment Group	Reference field used to identify the responsible group.
Date of Enquiry	Stores the enquiry date.

9.1. Field Design Considerations: -

★ Data Type Selection

- Choose appropriate data types (e.g., *String*, *Integer*, *Date/Time*, *Reference*) based on the nature of each field.

- Example: *Request Number* → Integer; *Date of Enquiry* → Date; *Customer Address* → String.

★ **Field Length and Constraints**

- Define maximum lengths for text fields to prevent overflow.
- Apply constraints like *mandatory*, *unique*, or *read-only* where necessary.

★ **Reference and Relationship Fields**

- Use reference fields (e.g., *Assigned To*, *Assignment Group*) to link related tables and maintain relational integrity.
- Ensure foreign keys are properly indexed for faster lookups.

★ **Validation Rules**

- Implement validation to ensure correct data entry (e.g., valid email format, non-empty mandatory fields).
- Use UI policies or scripts to enforce conditional visibility or mandatory logic.

★ **Default Values and Auto-Population**

- Pre-fill fields like *Work Status* or *Date of Enquiry* to streamline data entry.
- Automate repetitive entries using system defaults or workflow triggers.

★ **Security and Access Control**

- Restrict sensitive fields (e.g., *Customer Document*) to authorized roles.
- Apply field-level encryption or masking if required.

★ **User Interface and Labeling**

- Use clear, descriptive labels for each field to improve user understanding.
- Group related fields logically (e.g., Customer Info, Request Details).

★ **8. Performance Optimization**

- Index frequently queried fields (*Request Number*, *Work Status*) for faster retrieval.
- Avoid unnecessary computed fields that slow down form loading.

10.1. Creation of Relationship: -

- Click All in the Application Navigator.
- Go to System Definition → Relationships.
- Click New.
- Enter Name as Approval Request.
- Set Applies to the table as a Network Database Table.
- Set Queries from table to the approval table used in the project.
- Keep the relationship Active.
- Configure the query/script as required.
- Click Submit / Update.

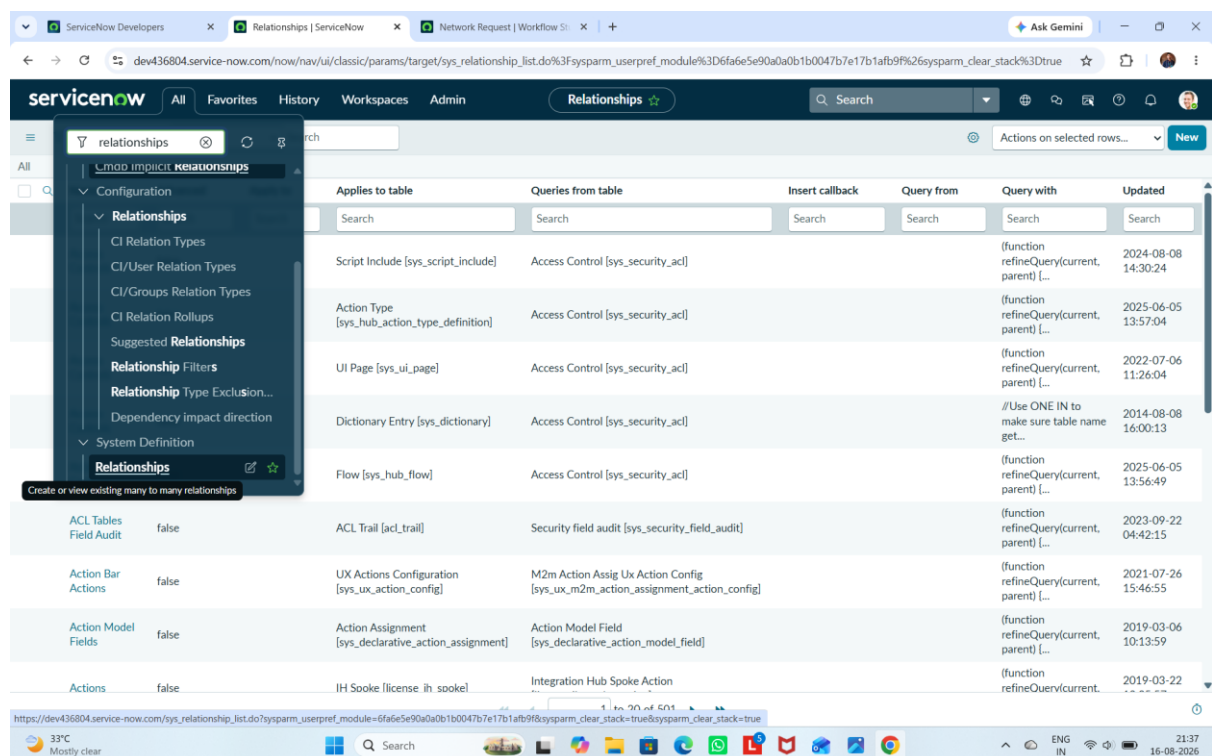


Fig.12: Creation of Relationships

10.2. Creation Of Related List: -

- Navigate to System Definition > Relationships.
- Click New to create a new relationship.
- Enter the relationship name as Approval Request.
- Set the Applies to the table to the Network Database Table.
- Set the Queries from table to Approval [sysapproval_approver]. Keep the relationship active. Save the relationship.

11.1. Relationship Code: -

- Open the Approval Request relationship.
- Locate the Query with / Query Script section.
- Enter the relationship query shown in the screenshot.
- Verify that the query uses the current Network Database Table record.
- Check the script for errors.
- Click Update.
- Verify the approval records through the related list.

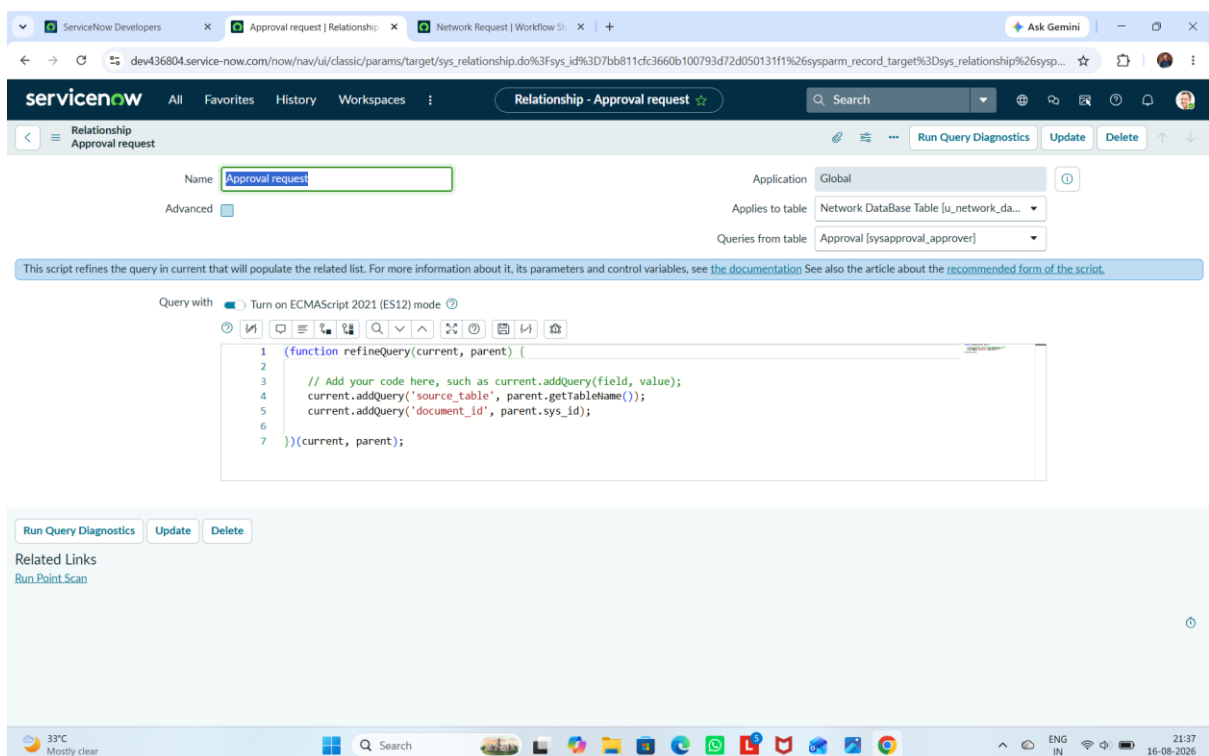


Fig.13: Script in Relationship

11.2. Output

The relationship provides a clear connection between the network request record and its approval information. This improves traceability because approval details can be reviewed in the context of the corresponding request.

12.1. Creation of Flow

- Click All.
- Go to Process Automation → Flow Designer.
- Open the Network Request flow.

- If it does not exist, click New.
- Enter the flow name and required details.
- Configure the trigger.
- Add the required actions in sequence.
- Save the flow.

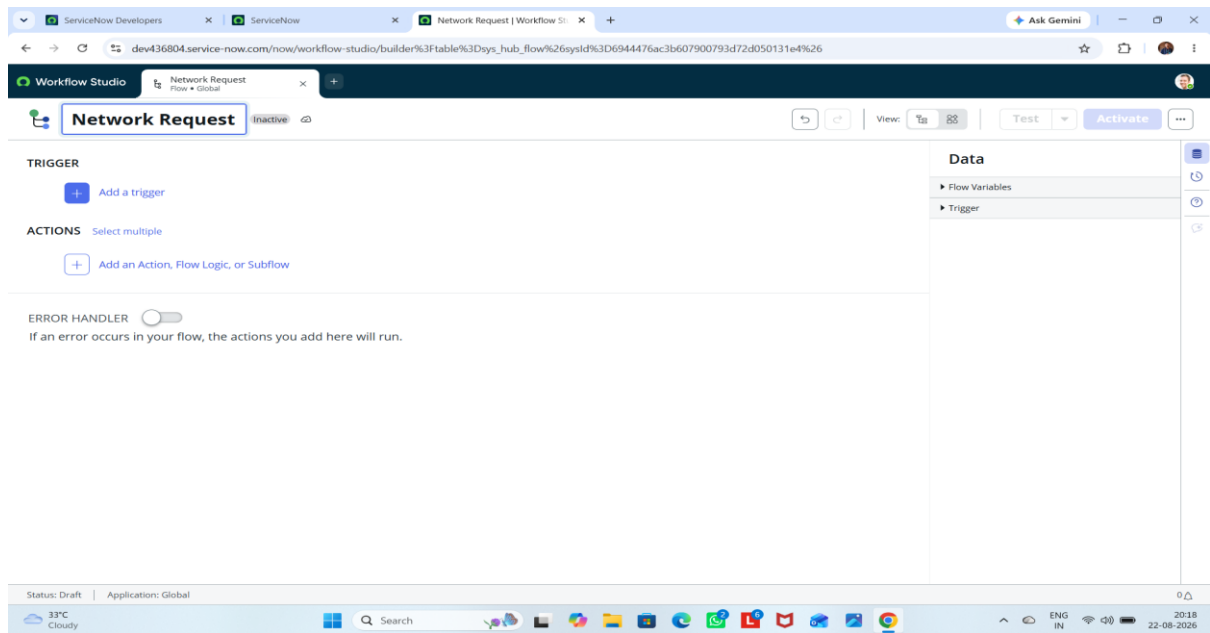


Fig.14: Flow Creation

12.2. Get Catalog Variables Creation: -

- Open the Network Request flow.
- Click Add Trigger.
- Select Service Catalog.
- Configure the Network Request / Requested Item condition.
- Review the trigger.
- Click Save.

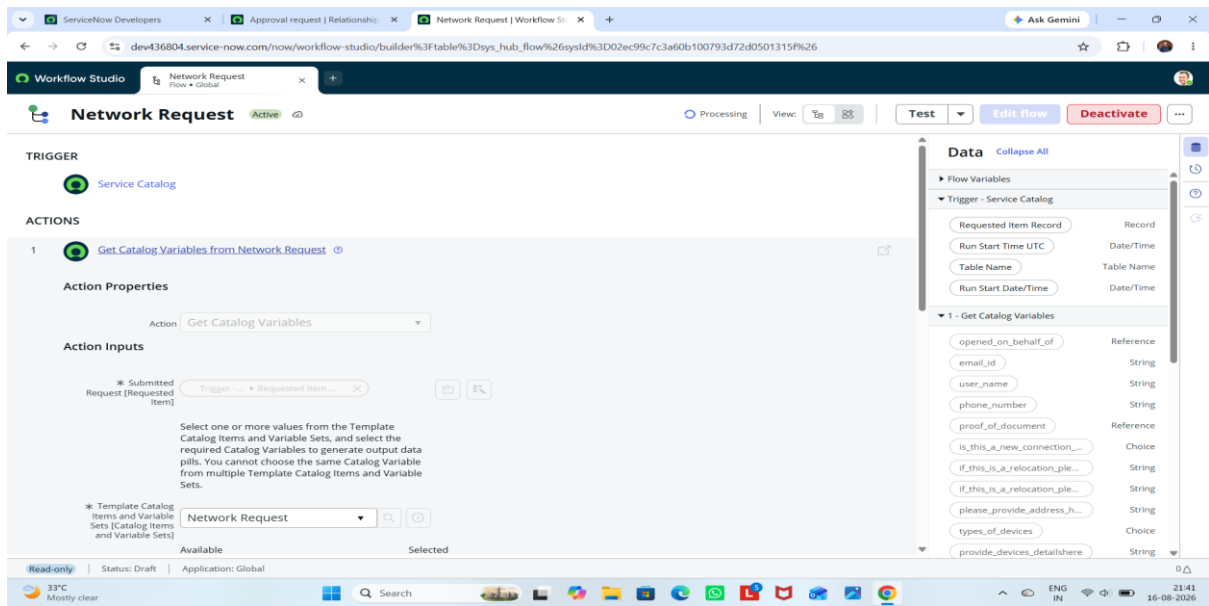


Fig.15: Creation of Get Catalog Variable from Network Request

- Click Add Action.
- Search for Get Catalog Variables.
- Select the action.
- Provide Requested Item from the trigger.
- Select the required catalog variables.
- Click Save.

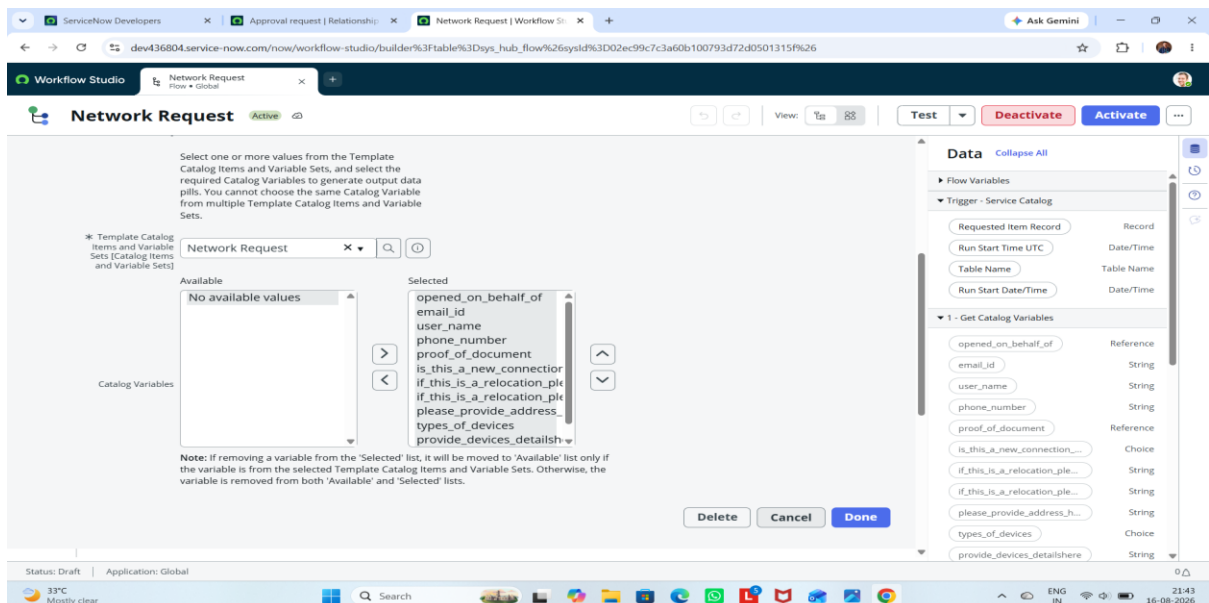


Fig.16: Get Catalog Variables Action

13.1. Record Creation: -

- Click Add Action.
- Select Create Record.
- Select Network Database Table.
- Map the catalog variable data pills to the correct table fields.
- Review every field mapping.
- Click Save.

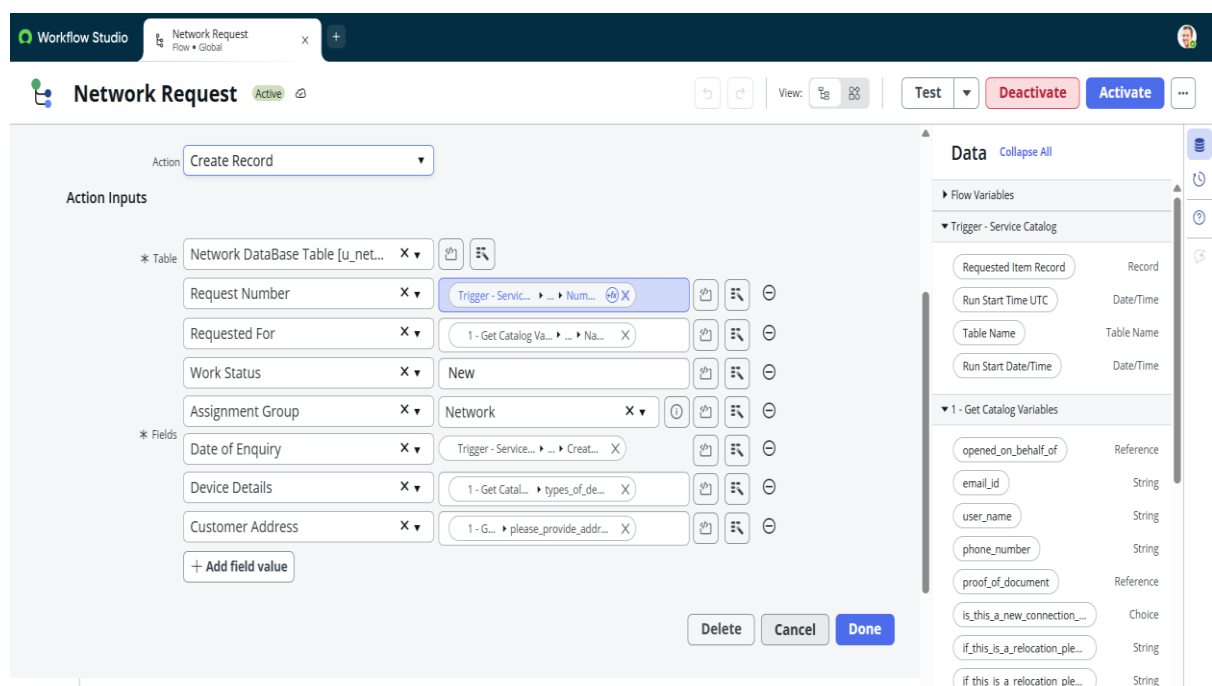


Fig.15: Record Creation

13.2. Ask for Approval Creation: -

- Add the Approval action after Create Record.
- Configure the approval details.
- Click Add Flow Logic.
- Select If.
- Configure the condition using the approval result.
- Save the condition.

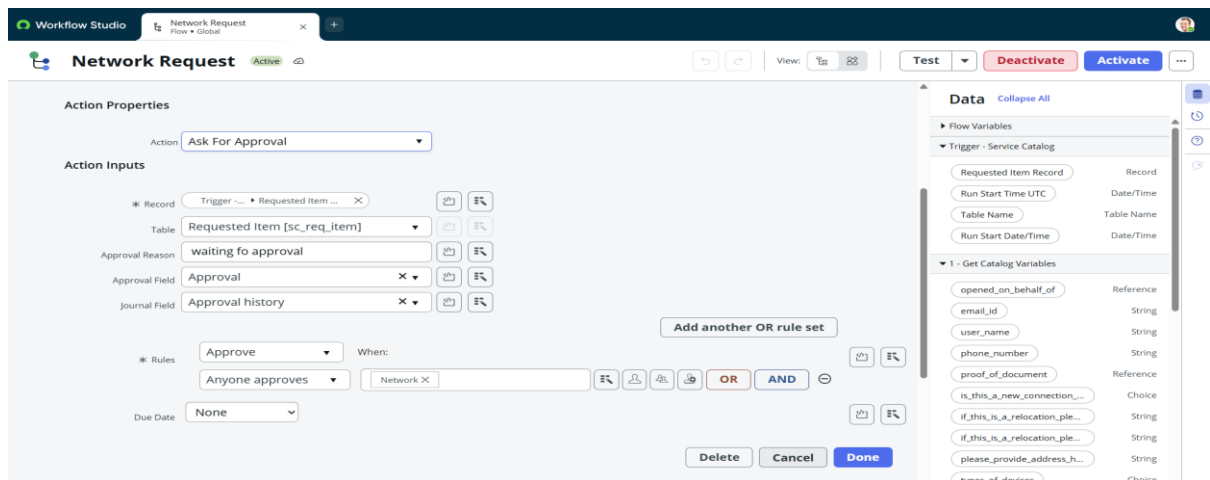


Fig.16: Creation of Ask for Approval

13.3. Send Email: -

- Open the required approval path.
- Click Add Action.
- Select Send Email.
- Configure the recipient.
- Enter the email subject.
- Enter the email message.
- Insert request information using data pills.
- Click Save.

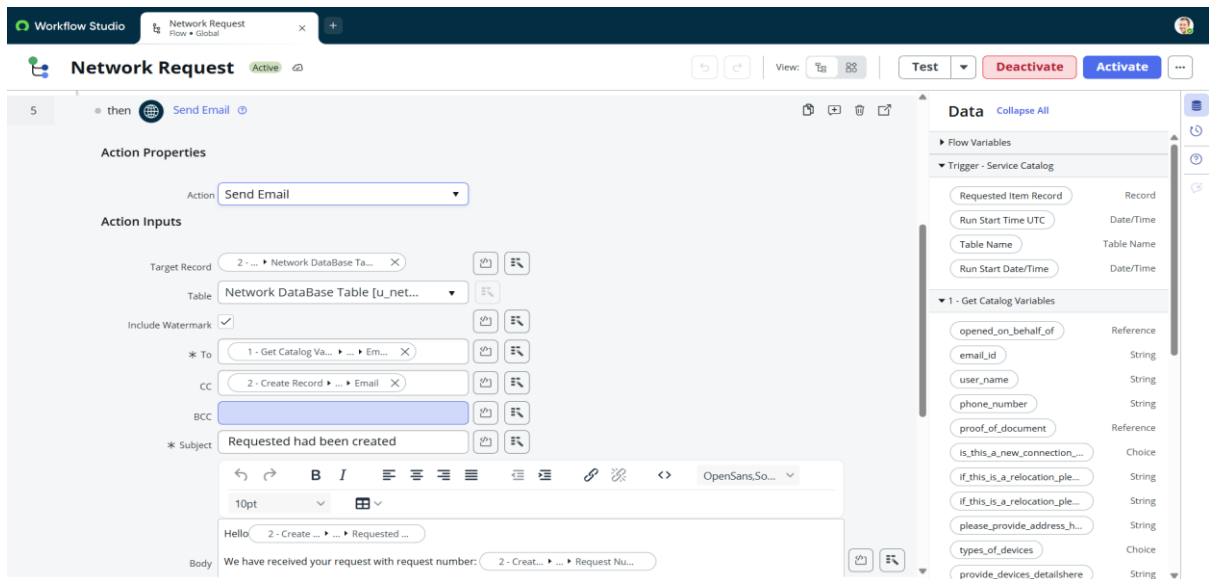


Fig.17: Send Email

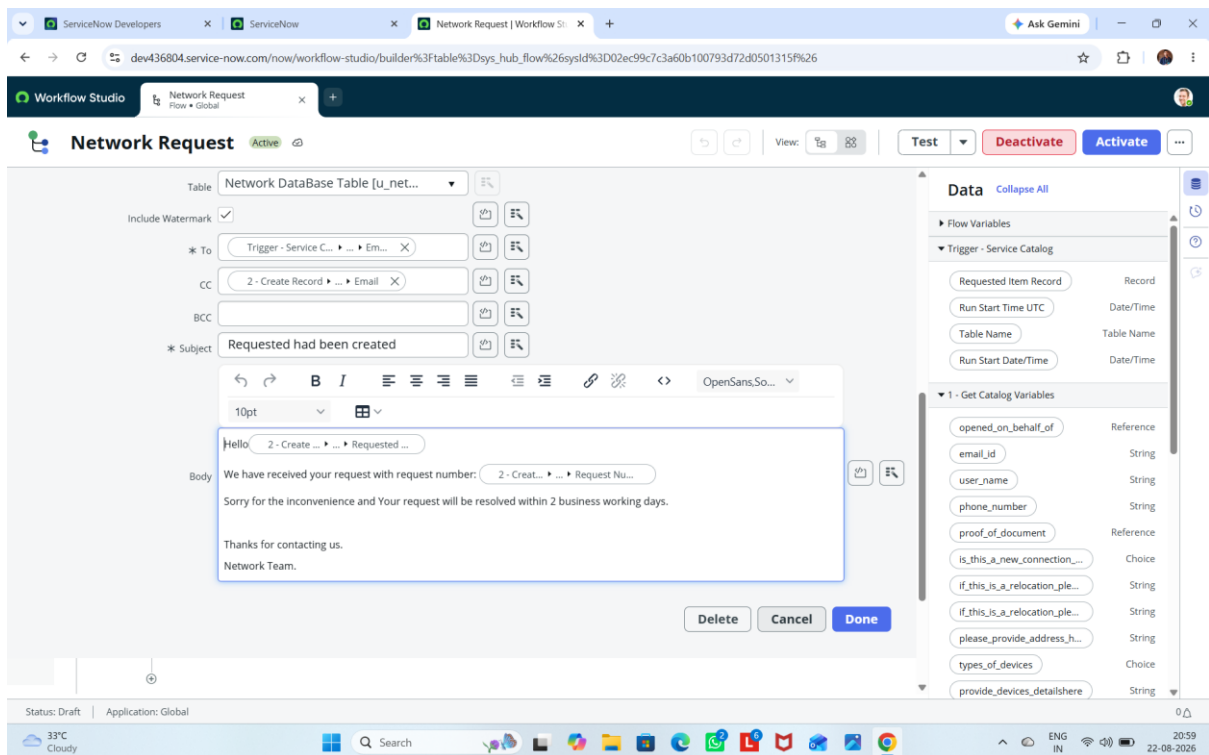


Fig.18: request Data Mapping

14.1. Update Record: -

- Click Add Action.
- Select Update Record.
- Select the Network Database Table record created earlier.
- Map the fields that must be updated.
- Review the values.

- Click Save.

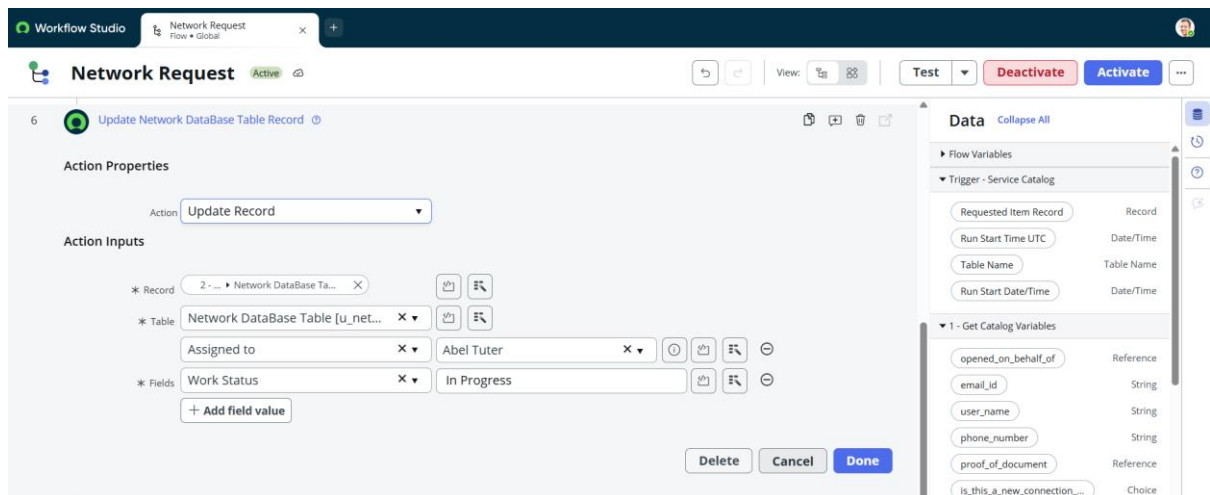


Fig.19: Record Update

14.2. Conclusion: -

- Verify Service Catalog Trigger.
- Verify Get Catalog Variables.
- Verify Create Record.
- Verify Approval / Condition.
- Verify Send Email.
- Verify Update Record.
- Check every data-pill mapping.
- Click Save.
- Activate the flow after final verification.

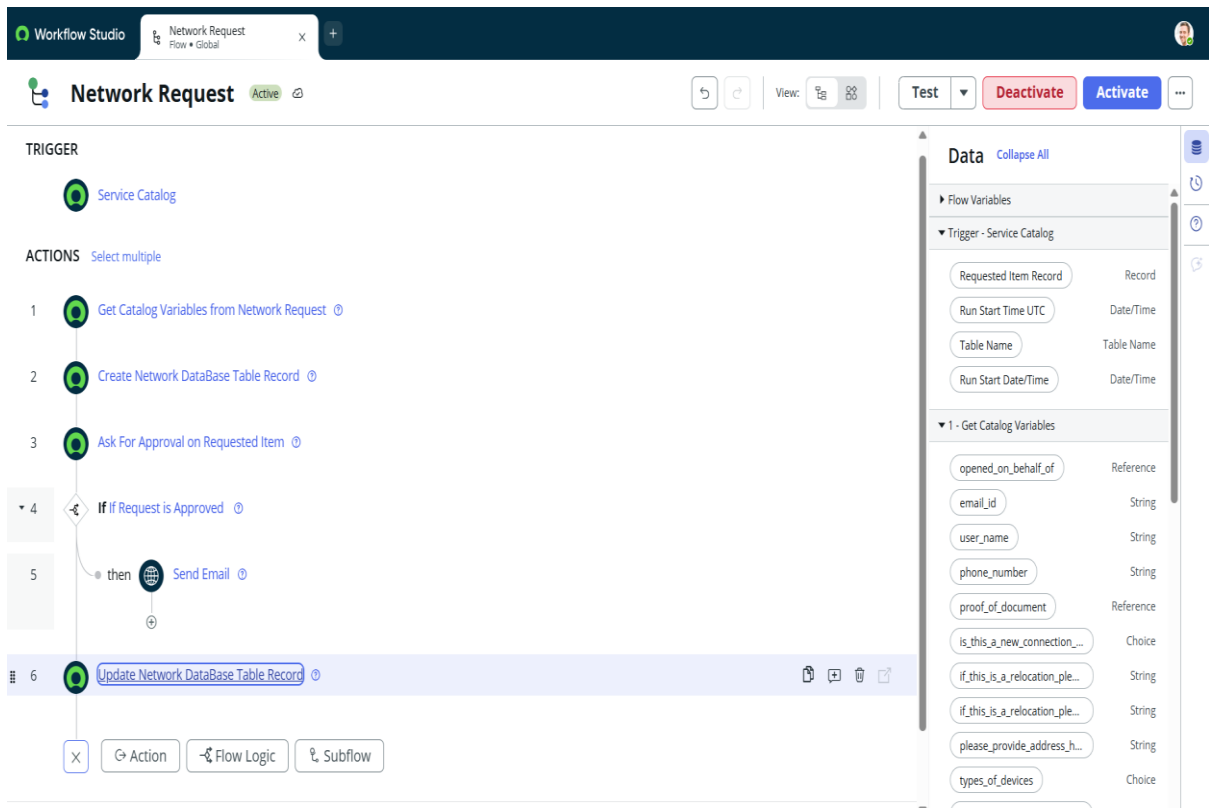


Fig.20: completion of Flow

14.3. Catalog Form Output: -

The catalog form represents the user-facing request interface and shows the fields that are completed during submission.

The screenshot shows the ServiceNow Network Request Catalog Form. The form is titled 'Network request Management' and is located under 'Service Catalog > Network and connectivity > Network Request'. The form contains several input fields: 'Opened on behalf of', 'Email Id', 'User name', 'Phone Number', 'Proof of Document' (with a 'Click to add...' link), and 'Is this a New connection or Relocation?' (with radio buttons for 'New', 'Relocation', and 'None'). There is also a field for 'If this is a relocation, Please provide...'. On the right side, there is a summary section titled 'Order this Item' which includes 'Quantity' (set to 1), 'Delivery time' (2 Days), and buttons for 'Order Now' and 'Add to Cart'. Below this, there is a 'Shopping Cart' section showing 'Empty'. The bottom of the screen shows a Windows taskbar with the date and time as 8:52 PM on 8/15/2026.

Fig.21: Network Request Catalog Form

ServiceNow Developers | Network Request | ServiceNow | Network Request | Workflow St... | Ask Gemini

dev436804.service-now.com/now/nav/ui/classic/params/target/com.glideapp.servicecatalog_cat_item_view.do%3Fv%3D1%26sysparm_id%3D12c9a84fc3eec7100793d72d0501317a

servicenow All Favorites History Workspaces Admin Network Request Search

Is this a new connection or relocation?

☒ New
☐ Relocation
☐ None

If this is a relocation, Please provide...

If this is a relocation, Please provide...1

Please provide address here

Types of Devices

☒ Laptop
☐ Mobiles
☐ Others

Provide devices detailshere

If anything else, please specify

33°C Partly cloudy Search 8:52 PM 8/15/2026

Fig.22: Network request Catalog Form completed

15.1 Testing the Flow: -

- Go to All → Process Automation → Flow Designer.
- Open the Network Request flow.
- Click Test.
- Select the required Requested Item test record.
- Start the test.
- Wait for the flow execution to complete.
- Open the execution details.
- Check each action for successful completion.

15.2. Execution of Test: -

- Open the completed Network Request flow.
- Use the Test option to execute the flow with a suitable request context.
- Allow the configured actions to run in sequence.
- Review the execution details for each action.
- Confirm that no action reports an unexpected failure.

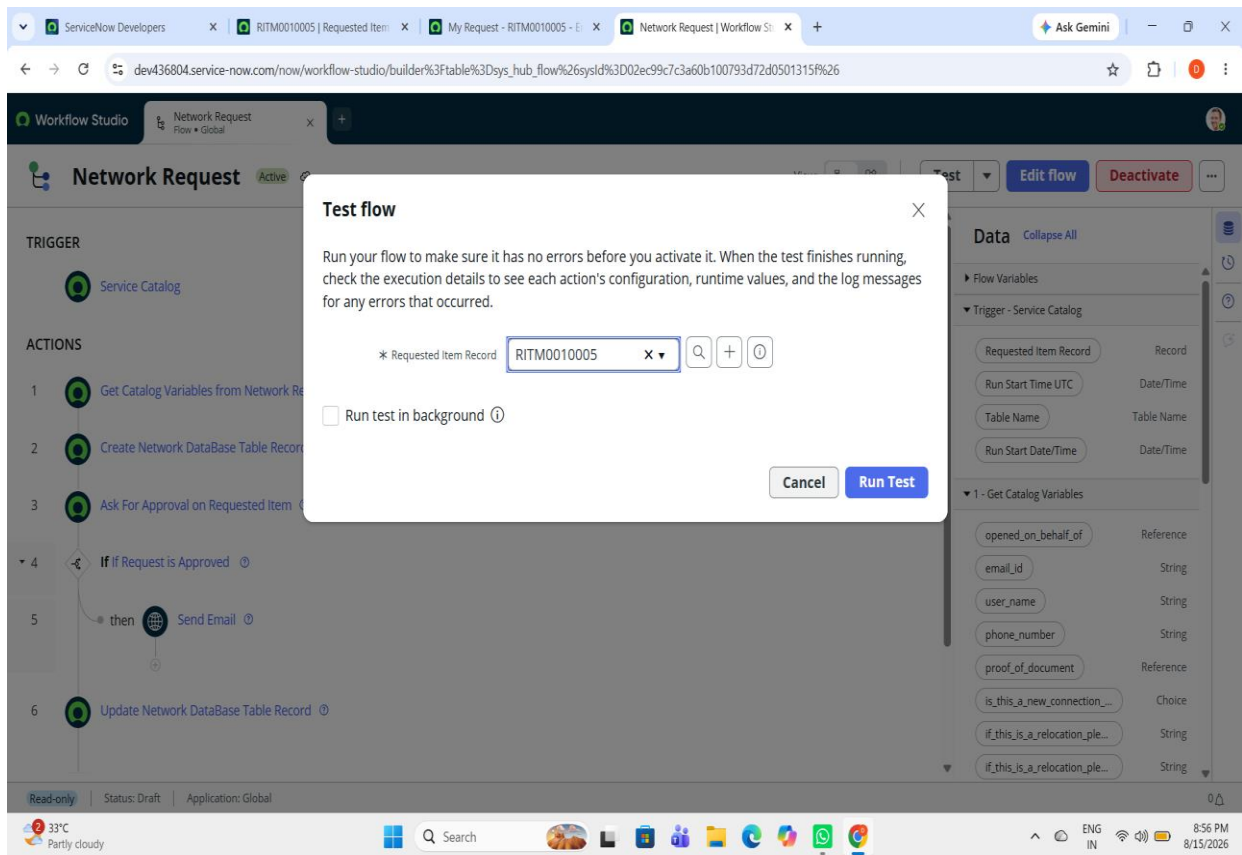


Fig.23: Test the Flow

15.3. Execution Output: -

The execution result provides evidence that the flow has been run and allows the developer to inspect the processing path and action outcomes.

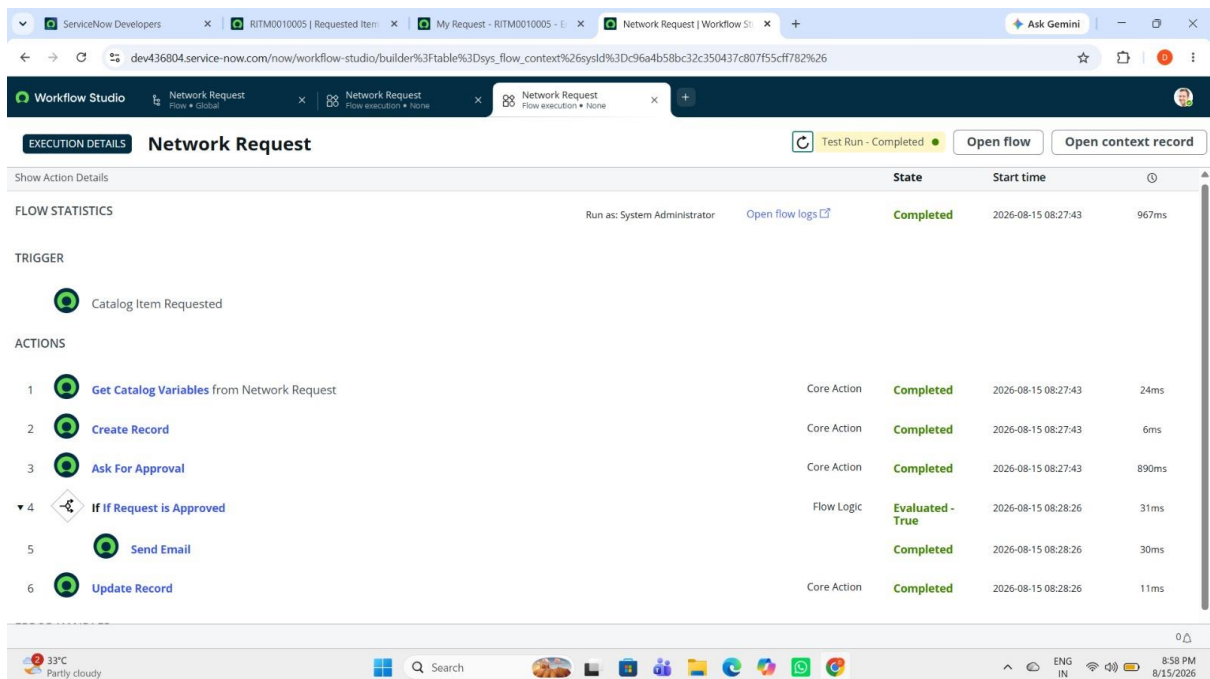


Fig.24: Flow Details-Test is completed

16.1. Notification Result: -

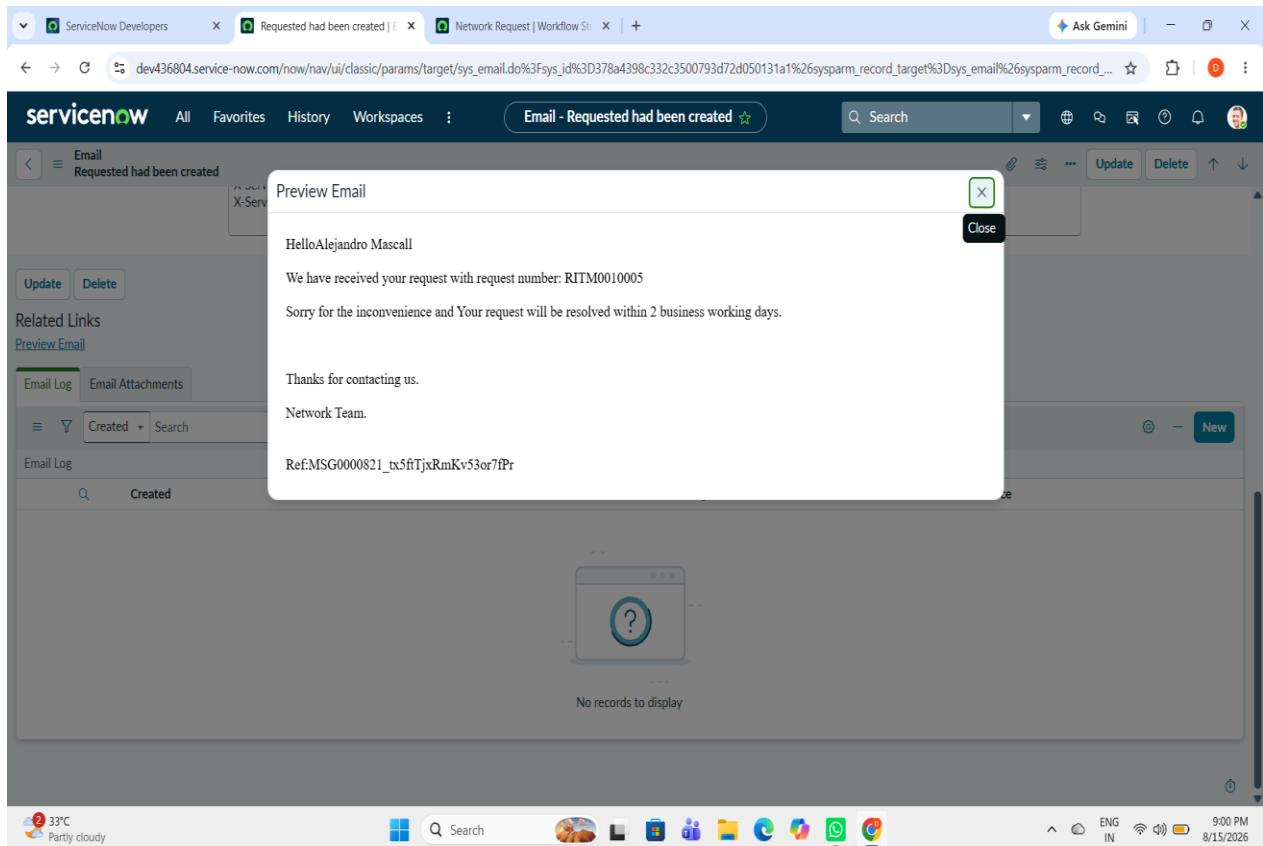


Fig.25: Notification Output

16.2. Conclusion: -

The *Automated Network Request Management System* developed on the ServiceNow platform successfully streamlines the process of handling network-related service requests. By integrating workflow automation, approval mechanisms, and real-time tracking, the system minimizes manual intervention and enhances operational efficiency. It ensures faster resolution times, improved transparency, and better communication between users and IT teams. Overall, this project demonstrates how ServiceNow's automation capabilities can transform traditional network request handling into a more reliable, scalable, and user-friendly process — contributing to higher productivity and service quality within the organization.

- Users receive a structured Network Request catalog form.
- Ten request-specific variables capture the required network information.
- Requester information is organized through a reusable Variable Set containing five variables.
- Two Catalog UI Policies provide dynamic form behavior.
- Network database Table fields provide a dedicated backend structure.
- Flow Designer automates data retrieval, record creation, approval processing, email notification, and record update.
- Testing and output screenshots provide evidence of the implemented process.

✓ The *Automated Network Request Management System* in ServiceNow effectively revolutionizes how network-related service requests are handled within an organization.

✓ By integrating automation, workflow orchestration, and real-time notifications, the system eliminates manual bottlenecks and ensures faster, more accurate request processing. It enhances collaboration between technical teams and end users, improves transparency through centralized tracking, and supports data-driven decision-making.

✓ Overall, this project demonstrates the power of ServiceNow's automation framework in achieving operational excellence — delivering a scalable, efficient, and user-centric solution that aligns with modern IT service management standards.